PYTHON FOR LOOP — COMPLETE NOTES

Beginner-friendly notes with important concepts and 10 practical examples.

1. What is a loop?

A loop is used to execute a block of code repeatedly. Python mainly provides two loops: for loop and while loop.

2. What is a for loop?

A for loop is used when you want to iterate over the items of a sequence or iterable such as a list, tuple, string, set, dictionary, or range.

3. Basic syntax

```
for variable in iterable:
    # statements
```

Example:

```
for i in range(5):
    print(i)

Output:
0
1
2
3
4
```

4. Important points

• Python uses indentation to define the loop body.
• The loop variable receives one item at a time.
• A for loop can iterate over strings, lists, tuples, sets, dictionaries, and ranges.
• range() is commonly used for number-based repetition.
• break stops the loop immediately.
• continue skips the current iteration and moves to the next one.
• pass does nothing and is used as a placeholder.

5. range() function

```
range(stop)
range(start, stop)
range(start, stop, step)
```

The stop value is excluded. The default step is 1.

```
range(5)          # 0,1,2,3,4
range(2, 6)       # 2,3,4,5
range(2, 10, 2)   # 2,4,6,8
range(10, 0, -2)  # 10,8,6,4,2
```

6. for loop with string

```
for ch in "Python":
    print(ch)
```

7. for loop with list

```
names = ["Akshay", "Rahul", "Amit"]
```

```
for name in names:
    print(name)
```

8. break, continue and pass

```
for i in range(1, 6):
    if i == 4:
        break
    print(i)

for i in range(1, 6):
    if i == 3:
        continue
    print(i)

for i in range(3):
    pass
```

FOR LOOP — 10 EXAMPLES

Example 1: Print numbers 1 to 10

```
for i in range(1, 11):
    print(i)
```

Example 2: Print even numbers from 1 to 20

```
for i in range(2, 21, 2):
    print(i)
```

Example 3: Print odd numbers from 1 to 20

```
for i in range(1, 21, 2):
    print(i)
```

Example 4: Sum of numbers 1 to 10

```
total = 0
for i in range(1, 11):
    total = total + i
print("Sum =", total)
```

Example 5: Multiplication table

```
n = 5
for i in range(1, 11):
    print(n, "x", i, "=", n * i)
```

Example 6: Count vowels in a string

```
text = "python programming"
count = 0

for ch in text:
    if ch in "aeiou":
        count += 1

print("Vowels =", count)
```

Example 7: Find the largest number in a list

```
numbers = [10, 45, 23, 89, 12]
largest = numbers[0]

for n in numbers:
    if n > largest:
        largest = n

print("Largest =", largest)
```

Example 8: Factorial of a number

```
n = 5
fact = 1

for i in range(1, n + 1):
    fact *= i

print("Factorial =", fact)
```

Example 9: Reverse a string

```
text = "Python"
reverse = ""
```

```
for ch in text:
    reverse = ch + reverse

print(reverse)
```

Example 10: Check prime number

```
n = 17
count = 0

for i in range(1, n + 1):
    if n % i == 0:
        count += 1

if count == 2:
    print("Prime number")
else:
    print("Not a prime number")
```